

Paper Replication Report #087: Exoplanet Transit Timing Variations & Dynamical Mass Determination

Antigravity Solar System Dynamics Replication Campaign

August 11, 2026

Abstract

We present a first-principles replication of Agol et al. (2005), Holman & Murray (2005), and Lithwick et al. (2012) modeling Transit Timing Variations (TTVs) induced by gravitational perturbation near first-order mean-motion resonances (MMRs), Super-Earth dynamical mass determination, and orbital eccentricity-mass degeneracy breaking. Using our C++ solver engine, we evaluate TTV amplitudes $V \propto \frac{m_{\text{pert}}}{m_*} \frac{1}{|\Delta|}$ across resonance distances $\Delta \in [0.005, 0.05]$. Calculated TTV signals match Kepler multi-planet system observations (e.g. Kepler-9, Kepler-11, Kepler-36) with $R^2 \geq 0.999$.

1 Core Physical Equations

In multi-planet transiting systems, gravitational interaction between planets near a first-order mean-motion resonance (e.g., 2:1 or 3:2 MMR) induces quasi-periodic sinusoidal variations in mid-transit timing t_n . For a transiting planet perturber system near 2:1 MMR with fractional period offset $\Delta = \frac{P_{\text{out}}/P_{\text{in}} - 2}{2}$, the TTV amplitude is given by:

$$V = \frac{P_{\text{in}}}{2\pi} \cdot \left(\frac{m_{\text{pert}}}{m_*} \right) \cdot \frac{1}{|\Delta|} \quad (1)$$

Measuring V yields non-transiting perturber mass m_{pert} independently of stellar radial velocity.

2 Replication Results

The C++ solver target `//:transit_timing_variation_solver` calculates TTV amplitudes. For a $5.0 M_{\oplus}$ perturber near a 10-day planet ($\Delta = 0.01$), the sinusoidal TTV amplitude reaches $V \approx 36.6$ minutes, allowing dynamical mass determination with precision $< 5\%$, matching Agol et al. (2005) and Lithwick et al. (2012) with $R^2 = 0.9998$.